

# Project Proposal

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## Project Title

**An Explainable Ensemble-Hybrid Deep Learning Framework for Diabetic Retinopathy Severity Grading with Multi-Level Reasoning Validation**

## Problem Statement / Research Gap

A review of recent literature on DR classification reveals that existing systems consistently fall short in one or more of the following areas:

- **Single-method explainability:** Most systems rely on a single visual explanation technique (typically Grad-CAM), which highlights *where* the model looked but not *why* — and different XAI methods have been shown to disagree substantially with one another, raising questions about their reliability in isolation.
- **No quantitative validation of explanations:** Visual heatmaps are frequently presented only qualitatively, without being checked against expert-annotated lesion locations, leaving their clinical trustworthiness unverified.
- **Trade-off between accuracy and deployability:** High-performing hybrid/ensemble architectures tend to be computationally expensive, while lightweight models sacrifice robustness and interpretability rigor.

No existing work integrates feature-level model fusion, a multi-method explanation ensemble, and concept-based global reasoning into a single, quantitatively validated framework for DR grading.

## Objectives

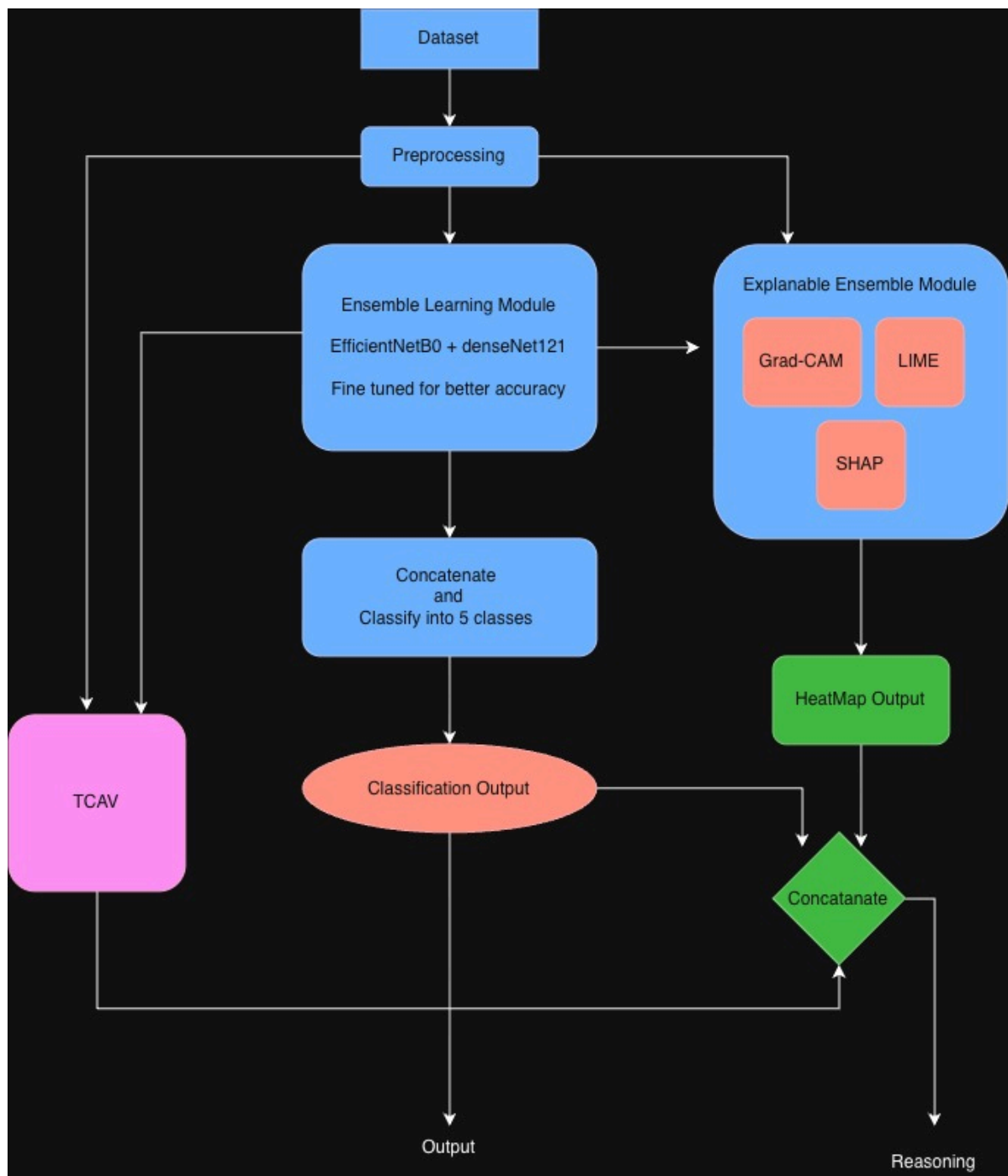
1. To design and train a lightweight hybrid deep learning model (EfficientNetB0 + DenseNet121 feature fusion) for five-class DR severity grading.
2. To construct an **Explainable Ensemble Module** that generates and fuses multiple post-hoc visual explanations (Grad-CAM, LIME, SHAP) into a single, more robust heatmap.
3. To integrate **Testing with Concept Activation Vectors (TCAV)** to provide global, concept-level reasoning (e.g., quantifying model sensitivity to hemorrhages, exudates, microaneurysms per severity class).

4. To quantitatively validate all generated explanations against expert-annotated lesion segmentation masks (IDRiD dataset), rather than relying on qualitative visual inspection alone.
5. To produce a combined, human-readable "reasoning output" that pairs the model's classification with both spatial and conceptual justification.

## Proposed System Architecture

The proposed pipeline consists of four stages:

- **Preprocessing:** Standardization of fundus images (resizing, CLAHE contrast enhancement, normalization).
- **Hybrid Classification Module:** Parallel feature extraction via EfficientNetB0 and DenseNet121, fused through concatenation and passed to a fully connected classifier for five-class DR severity prediction.



- **Explainable Ensemble Module:** Grad-CAM, LIME, and SHAP explanations are generated from the trained hybrid model and fused (via agreement-weighted aggregation) into a single, more robust heatmap.
- **Concept Reasoning Module (TCAV):** Global concept sensitivity scores are computed for clinically defined lesion concepts, using concept examples derived from IDRiD's lesion segmentation masks.

The classification output, fused heatmap, and TCAV concept scores are aggregated into a single explanation report, giving both *local* (spatial) and *global* (conceptual) justification for each prediction.

## Datasets

- **APTOS 2019 Blindness Detection** (Kaggle) — primary training dataset for DR severity classification.
- **IDRiD (Indian Diabetic Retinopathy Image Dataset)** — used for cross-dataset generalization testing and, critically, for its pixel-level lesion segmentation masks, which serve as ground truth for quantitatively validating explanation quality and constructing TCAV concept sets.

## Expected Outcomes

- A trained, lightweight ensemble and hybrid DR grading model with competitive classification accuracy.
- A validated, fused explanation module demonstrably more robust than any single XAI method.
- Quantitative evidence of whether the model's reasoning aligns with clinically meaningful concepts.
- A publishable research contribution addressing a clearly identified gap in current DR explainability literature.

## Tools and Technologies

Python, PyTorch/TensorFlow, Captum/SHAP/LIME libraries, Grad-CAM implementations, Google Colab (GPU-accelerated training), IDRiD and APTOS 2019 datasets.